

Reading, Not Asking: Probing a Small Language Model’s Signal for Selective Refusal in Grounded Question Answering

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Abstract—A grounded language model should answer only when the supplied evidence supports an answer, and refuse otherwise. RefusalBench measures this selective refusal and reports that small models rarely identify the correct reason for refusing. We reproduce its single-document benchmark (RefusalBench-NQ, 1,600 examples) with Qwen1.5-7B-Chat on a free-tier GPU and test two extensions. First, asking the model to write an evidence-state diagnosis before answering fails: it declares the evidence clear in 95% of cases and almost never refuses, which is worse than the baseline. Second, we stop trusting the model’s text and read its internal signal in a single forward pass: the probability that its reply opens with a refusal code, a yes/no sufficiency probability, and a linear probe on its hidden state. All three separate refusal-required from answerable examples well above chance (AUROC 0.73, 0.75 and 0.80). Against the baseline judged by Gemini, however, thresholding these probabilities gives no clear gain (balanced accuracy 0.66–0.67 versus 0.69), and the supervised probe is only slightly higher (0.72; difference +0.03, 95% CI $[-0.01, +0.07]$). A small model’s knowledge of when to refuse is present in its states, but simple readouts do not yet exploit it reliably. Code and per-example scores are released.

Index Terms—selective refusal, grounded question answering, retrieval-augmented generation, linear probing, small language models

I. Introduction

Retrieval-augmented generation (RAG) [1] lets a language model answer from supplied passages instead of from memory. A grounded system can still fail even when retrieval succeeds: the passage may be ambiguous, contradict itself, omit the needed fact, rest on a false premise, be at the wrong level of detail, or ask for an opinion. In those cases the correct behavior is to refuse. RefusalBench [2] turns this into a benchmark with controlled perturbations and shows that even strong models cannot both answer well and refuse correctly; small open models are weaker still.

This paper asks a narrow question about one small open model, Qwen1.5-7B-Chat [3], [4]: can we improve its answer-or-refuse decisions without changing its weights, and where does the information needed for that decision live? We make three contributions.

- 1) We reproduce the RefusalBench-NQ baseline for Qwen1.5-7B-Chat on a free Colab T4 GPU with a free judge, documenting every deviation from the original setup.

- 2) We report a clear negative result: making the model write an evidence-state diagnosis before answering (“Classify-then-Decide”) collapses refusal behavior, with a format-only control to rule out a formatting explanation.
- 3) We show that the model’s internal signal for refusing is substantial (AUROC 0.73–0.80) but that simple readouts, a threshold on its refusal probability or a supervised probe, give at most a small and statistically unclear improvement over the baseline.

We frame this as a lightweight study of one model, not a general solution; Section VIII states its limits.

II. Related Work

Refusal and unanswerable questions. Reading-comprehension benchmarks introduced unanswerable questions so that models must abstain when the passage does not contain the answer [5]. RefusalBench [2] generalizes this to six controlled kinds of evidence failure and separates detection (should the model refuse?) from categorization (why?), finding the second much harder. We use its released NQ-based benchmark, built on Natural Questions [6].

What models know about themselves. Large language models can be well calibrated about whether they know an answer when asked in the right format [7]. Linear classifier probes [8] read information from intermediate layers; truthfulness signals have been recovered from hidden states with and without supervision [9], [10]. We apply the same idea to a different target: whether the evidence supports answering.

Selective prediction. A classifier can abstain when its confidence is below a threshold, trading coverage for risk [11], and confidence is only useful if calibrated [12]. Our threshold gate is selective prediction with the model’s own refusal probability as the confidence.

III. Task, Data and Metrics

Benchmark. RefusalBench-NQ has 1,600 examples derived from 100 Natural Questions source questions. Each example is a query with a passage. 536 are answerable (low-intensity perturbations) and 1,064 require a refusal, spread

over six classes: ambiguity, contradiction, missing information, false premise, granularity mismatch and epistemic mismatch. The model must reply with an answer or one of seven refusal codes (REFUSE_AMBIGUOUS_QUERY, ..., REFUSE_OTHER). Because the 1,600 examples come from only 100 source questions, they are not independent; all our intervals therefore resample whole source questions (cluster bootstrap [13], [14]).

Metrics. Following the benchmark, answer accuracy is the share of answerable examples answered correctly (judge score ≥ 4 of 5), refusal accuracy is the share of refusal-required examples refused with the exactly correct code, the false refusal rate (FRR) is the share of answerable examples refused, and the missed refusal rate (MRR) is the share of refusal-required examples answered. For answer-or-refuse detection we also report

$$\text{BA} = 1 - \frac{1}{2}(\text{FRR} + \text{MRR}), \quad (1)$$

the balanced accuracy, where 0.5 is chance, and the F1 score of the “refuse” class. BA weights both error types equally, which matters because two thirds of the examples require a refusal.

IV. Baseline Reproduction

Setup. We run Qwen1.5-7B-Chat with Hugging Face Transformers [15] in 4-bit NF4 quantization [16] on a Colab T4, using the benchmark’s prompt verbatim, temperature 1.0, top- p 1.0, at most 256 new tokens, seed 0. The original work served the model with vLLM on A100 GPUs and judged replies with Claude Sonnet 4. We replace the judge by Gemini (gemini-3.5-flash-lite, temperature 0) [17], with parse-first scoring: a reply containing exactly one clean refusal code is scored by code, every other reply goes to the judge. Generation succeeded for 1,560 of 1,600 examples; 40 failed with GPU out-of-memory errors and are excluded (2.5%, slightly longer inputs than average).

Reply format. Only 528 of the 1,560 replies (34%) are a clean refusal code; 991 (64%) are free text. The judge therefore carries most of the scoring, and any method that reads only the code understates the model’s refusals.

Results. Table I gives the baseline on the 770 replies graded so far (a random half; grading order is shuffled, and it is limited by the free judge quota) [TODO: update with all 1,560 graded replies]. Our answer accuracy is higher than the published 0.561 and refusal accuracy is low in both, as expected for a 7B model; the exact values differ because of the judge, quantization and inference-stack changes. The baseline misses most refusals for ambiguity (0.74) and contradiction (0.71) but few for false premise (0.19), granularity mismatch (0.25) and epistemic mismatch (0.30), and it over-refuses false-premise-class answerable examples (FRR 0.33).

V. Extension 1: Classify-then-Decide

Idea. If the model first names what is wrong with the evidence, its final decision should improve. The prompt

TABLE I

Baseline, Qwen1.5-7B-Chat on RefusalBench-NQ (770 graded replies: 509 refusal-required, 261 answerable). The paper values are those reported for Qwen1.5-7B; the refusal accuracy is read from a small plot.

Metric	Paper	Ours
Answer accuracy	0.561	0.690
Refusal accuracy (exact code)	≈ 0.05	0.114
False refusal rate	–	0.165
Missed refusal rate	–	0.456
Detection F1	–	0.665

TABLE II

Extension 1: the evidence state Qwen wrote, by true perturbation class (800 examples). “Other” is a valid non-CLEAR state; “none” is a missing or malformed state.

True class	n	CLEAR	Other	None
Answerable	268	258	0	10
Ambiguity	69	68	0	1
Contradiction	94	87	1	6
Missing information	92	87	0	5
False premise	93	89	2	2
Granularity mismatch	92	83	4	5
Epistemic mismatch	92	90	0	2

keeps the benchmark’s definitions and asks for two lines: EVIDENCE_STATE: (one of CLEAR or six states mapping to the refusal codes) and FINAL:. A non-CLEAR state determines the refusal code in software; CLEAR means the text after FINAL: is the answer.

Control. A format-only condition adds just the instruction to end with a FINAL: line, so any gain could be attributed to cleaner output rather than to diagnosis. All three conditions use the same model, settings and a stratified subset of 800 examples (268 answerable, 532 refusal-required, covering all 100 source questions); we compare them on the 776 examples that also exist in the baseline.

Result. The model answers EVIDENCE_STATE: CLEAR on 762 of 800 examples (95%), including 504 of the 532 refusal-required ones (Table II), and then answers normally. Because the state is read directly from the text, detection can be scored without a judge (Table III). Classify-then-Decide detects 1.9% of the required refusals, against 43.3% for the baseline and 35.8% for the format-only control (clean codes only, so the latter two are lower bounds). The correct-category rate falls from 6.8% (baseline) and 3.3% (control) to 0.0%; the differences to both are significant (paired cluster bootstrap: -6.8 points, 95% CI $[-9.0, -4.5]$ versus the baseline; -3.3 points, $[-4.8, -1.9]$ versus the control). The control itself is slightly worse than the baseline, so formatting alone does not explain the collapse. Asking this model to commit to a diagnosis first anchors it on “clear” and makes it worse.

TABLE III

Extension 1 on 776 common examples, clean refusal codes only (no judge). Baseline and control are lower bounds because free-text refusals are not counted.

	Baseline	Format-only	Classify-then-Decide
Refusals detected (%)	43.3	35.8	1.9
Correct category (%)	6.8	3.3	0.0
False refusal (%)	15.4	7.7	0.4

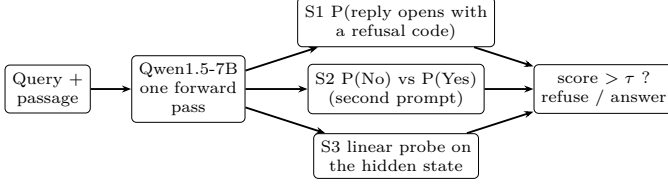


Fig. 1. Extension 2: no text is generated. One forward pass gives three scores; a threshold τ (or the probe) turns each into an answer-or-refuse decision, with τ chosen on other source questions.

VI. Extension 2: Reading the Refusal Signal

Idea. Instead of reading what the model writes, read how likely it is to refuse. We use one forward pass per example with no text generation and three scorers; higher always means “refuse”. A decision is “refuse iff the score exceeds a threshold τ ” (Fig. 1).

S1: refusal-code probability. With the original prompt, we compute the probability that the reply opens with a refusal code. A code spans several tokens and the model sometimes wraps it, so for each wrapper $w \in \{\text{none}, **, ', \text{ANSWER} :, \text{Answer} : \}$ we take the longest token prefix v_w shared by all seven codes and compute its probability by teacher forcing; the opening forms are mutually exclusive, so

$$p_{\text{ref}}(x) = \sum_w \prod_{t=1}^{|v_w|} P(v_{w,t} | x, v_{w,<t}), \quad s_1 = \log \frac{p_{\text{ref}}}{1 - p_{\text{ref}}}. \quad (2)$$

The default threshold is $p_{\text{ref}} = 0.5$ ($s_1 = 0$), what the model does on its own; the wrapped forms account for about 30% of the baseline’s coded refusals. The cached scoring was checked against a cache-free reference on a tiny random model (maximum difference 10^{-6}).

S2: sufficiency probability. A second prompt asks whether the query can be answered completely and faithfully from the passage, “Yes or No”; $s_2 = \log P(\text{No}) - \log P(\text{Yes})$ over the first token (mean probability mass on the two words: 1.0).

S3: hidden-state probe. A logistic regression on the last-token hidden state (final and middle layer, under either prompt), after standardization and PCA to 128 dimensions. It is supervised with the benchmark’s labels, unlike the baseline.

Protocol. All 100 source questions occur in any random half of the data, so tuning on one half and testing on

TABLE IV

Extension 2 on all 1,600 examples; thresholds and probe cross-fitted over source questions; 95% cluster-bootstrap intervals.

Scorer / rule	AUROC	FRR	MRR	BA	F1
S1 refusal prob.	.726				
default ($p > 0.5$)		.103	.590	.654	.561
tuned threshold		.256	.408	.668	.688
S2 yes/no prob.	.751				
default (No > Yes)		.125	.540	.667	.604
tuned threshold		.321	.329	.675	.732
S3 hidden-state probe	.797	.250	.307	.721	.762

the other would leak source questions. We instead cross-fit over source questions (5 folds): an example’s own source question is never in the data used to choose its threshold (maximizing BA) or to fit its probe, whose layer and regularization are chosen by an inner grouped cross-validation. Held-out decisions are pooled for evaluation. The success rule was fixed before the run: the signal check requires the lower end of S1’s AUROC interval to exceed 0.5, and the gate check requires the tuned-minus-default BA interval to lie above 0. Scoring takes about 0.3 s per example and prompt on a T4 (about 8 minutes per pass over the 1,600 examples), needs no judge and no API key, and costs nothing on the free tier.

A. Does the signal exist?

Table IV shows that all three scorers separate refusal-required from answerable examples far above chance on all 1,600 examples. The probe is best; S2 is slightly better than S1 (S1 minus S2: -0.025 , 95% CI $[-0.047, -0.003]$), and the probe beats S1 by $+0.071$ $[+0.041, +0.102]$. Figure 2 shows the S1 score for the two groups: they overlap but are clearly shifted. The probe chose a middle-layer state in four of the five folds (three under the baseline prompt, one under the yes/no prompt) and the final-layer state of the yes/no prompt in the fifth. Tuning a threshold on S1 moves the operating point (MRR 0.59 to 0.41, FRR 0.10 to 0.26) but does not significantly improve balanced accuracy over the default ($+0.015$, $[-0.008, +0.036]$): it slides along one trade-off curve. The probe is better than the tuned S1 gate by $+0.053$ $[+0.024, +0.081]$.

B. Is it better than the baseline?

The fair comparison scores the judged baseline replies and the new decision rules on the same examples (Table V, Fig. 3). A baseline reply counts as a refusal when the judge classifies it as any refusal code. On the 770 graded examples the baseline reaches BA 0.688 (95% CI $[0.654, 0.722]$), higher than the 0.64 obtained by counting only clean codes, which illustrates how much a code-only reading understates it. The label-free default thresholds do not beat it, and S1’s default is slightly worse (-0.029 , $[-0.056, -0.002]$). The tuned thresholds and S2 are indistinguishable from it. The probe is the only candidate: BA 0.718, a difference of $+0.030$ with

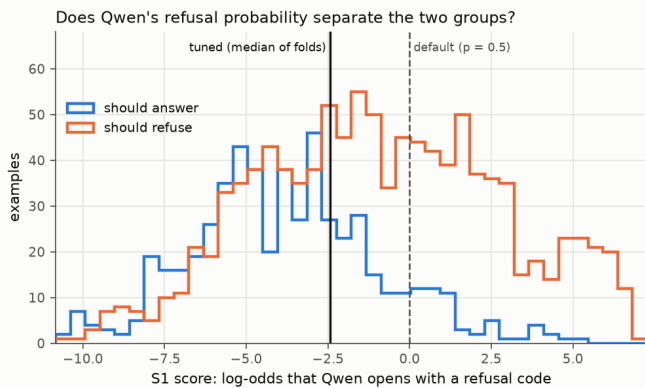


Fig. 2. S1 score (log-odds that the reply opens with a refusal code) for answerable and refusal-required examples, all 1,600 examples. The default threshold is $s_1 = 0$.

TABLE V

Judged baseline versus the new decision rules on the same 770 examples (BA difference is rule minus baseline, 95% cluster-bootstrap CI). Labels: whether the rule was fitted on benchmark labels.

Rule	BA	Δ BA [95% CI]	Labels	Verdict
Baseline (judged)	.688	—	no	—
S1 default	.658	-.029 [-.056, -.002]	no	worse
S1 tuned	.671	-.016 [-.045, +.011]	yes	unclear
S2 default	.669	-.018 [-.045, +.008]	no	unclear
S2 tuned	.668	-.019 [-.054, +.016]	yes	unclear
S3 probe	.718	+.030 [-.012, +.067]	yes	unclear

an interval that includes zero, $[-0.012, +0.067]$. It misses fewer refusals (MRR 0.29 versus 0.46) at the price of more false refusals (0.28 versus 0.17). Figure 3 shows the baseline’s operating point lying on the S1 and S2 curves and just above the probe’s: the model’s free-text decisions already use about as much of the signal as a threshold on its probability can recover. [TODO: re-run with all 1,560 baseline replies graded and update Table V, Fig. 3 and this paragraph]

C. Where does each rule help?

Table VI breaks the decisions down by perturbation class. The three readouts behave differently. S1 shares the baseline’s profile: it still misses 67% of ambiguity and 63% of contradiction refusals. S2 and the probe cut contradiction misses from 0.71 (baseline) to 0.26 and 0.22, and ambiguity misses from 0.74 to 0.54 and 0.47, but they pay with more false refusals on answerable examples of those classes (contradiction 0.17 to 0.44 and 0.30). The probe is the only rule that lowers the false-premise false-refusal rate (0.20 against 0.49–0.54 for S1 and S2) but it misses more false-premise refusals than the baseline (0.28 against 0.19) and over-refuses granularity-mismatch examples (0.38 against 0.10). The overall BA of the probe therefore hides large class-dependent trade-offs.

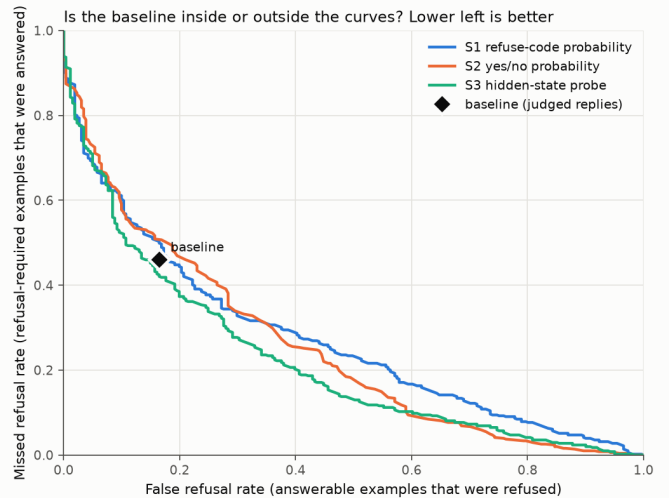


Fig. 3. False versus missed refusal for every threshold of S1, S2 and S3 (lower left is better), with the judged baseline (diamond) on the same 770 examples.

TABLE VI

Missed / false refusal rate by perturbation class. Baseline: 770 graded examples; other columns: all 1,600 examples at the thresholds that maximize overall BA. Lower is better in both entries.

Class	Baseline	S1 tuned	S2 tuned	S3 probe
Ambiguity	.74/.09	.67/.14	.54/.27	.47/.23
Contradiction	.71/.17	.63/.21	.26/.44	.22/.30
Missing information	.56/.22	.53/.27	.58/.25	.56/.21
Epistemic mismatch	.30/.12	.22/.13	.26/.14	.13/.17
False premise	.19/.33	.22/.54	.17/.49	.28/.20
Granularity mismatch	.25/.10	.23/.20	.22/.30	.23/.38

VII. Discussion

Asking fails, reading works somewhat. Prompting a 7B model to reason about the evidence before acting produced its worst behavior, while the same model’s probabilities and hidden states already separate the two groups with AUROC up to 0.80. The information about whether to refuse is present; the failure is in what the model is willing to write. We did not test why the model anchors on “clear”, so we make no mechanistic claim.

Where it works and where it does not. All methods catch epistemic mismatch, false premise and granularity mismatch reasonably well (MRR about 0.13–0.30). Ambiguity, contradiction and missing information are the hard classes: S1 and the baseline miss most of them, whereas S2 and the probe recover many contradictions and ambiguities, but only by refusing more answerable examples. The false-premise class is over-refused by S1 and S2 (FRR 0.54 and 0.49). Because the best threshold differs by class, a single global threshold is a blunt instrument; whether class-aware thresholds would help is an open question we did not test, and with one seed and 100 source questions the per-class numbers are descriptive, not tested.

What this suggests. Reading a refusal probability through a threshold is not an improvement over letting the model answer; a supervised probe is a plausible but small one. Our result is compatible with the view that selective refusal in small models is limited by how the knowledge is expressed rather than by its absence, but it does not establish it.

VIII. Limitations

(i) One model, one benchmark, one seed, 4-bit weights, sampling at temperature 1.0. (ii) The judge is Gemini, not the original Claude, and has not yet been validated by our planned 30-reply hand check. (iii) The baseline comparison currently uses half of the graded replies, and the 40 failed examples are excluded. (iv) The probe is supervised and trained on the same 100 source questions’ distribution; cross-fitting prevents source overlap but not distribution shift. (v) S1 reads only openings that start with a refusal code, so free-text refusals look like answers to it. (vi) We measure detection, not the reason for refusing and not answer correctness for the new rules. (vii) We compare five rules with 95% intervals and no multiplicity correction, so the one “worse” verdict (S1 default, upper bound -0.002) is not robust to such a correction.

IX. Conclusion

We reproduced the RefusalBench-NQ baseline for Qwen1.5-7B-Chat on free resources, showed that making the model diagnose the evidence before answering backfires, and showed that its internal refusal signal is strong (AUROC 0.73–0.80) yet only weakly exploitable: thresholding it does not beat the baseline and a supervised probe is at best slightly better. Future work should complete the judged baseline comparison, validate the judge, repeat the study on more models and seeds, and test readouts that target the hard classes, ambiguity and contradiction. The code, configurations and per-example scores are at <https://github.com/aidepartmentrah-pixel/refusalbench-reproduction>.

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